

DDTA - Scheda di lettura compilata

SRC-0041 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso	
Source ID	SRC-0041
Titolo	Kernel Modeling Language (KerML) Version 1.0
Autori/ente	Object Management Group (OMG)
Anno	2025
Identificatore	KerML 1.0; formal adoption September 2025; current normative PDF OMG file id formal/26-03-01
Accesso	public_normative_specification
Verification state	verified_full_text_public

1. Research problem and contribution

Defines a foundational modeling language with semantics for types, features, classes, structures, associations, behaviors, connectors, interactions and flows. It is intentionally more general than the systems-engineering concepts layered by SysML.

2. Starting artifacts and generated representations

The kernel distinguishes data values/types, structural classes, associations/links, behaviors/performances and features. A Structure is primarily structural and does not itself become Behavior, though it may perform or participate in behaviors. Connectors relate features; flows have source/target ends and payload; interactions combine behavior and association semantics.

3. Traceability, change, automation and human role

KerML primarily provides semantic foundations rather than a lifecycle/governance workflow. It gives precise identity/type/relationship semantics suitable for machine-readable interchange but does not itself solve document-to-model authority, human acceptance or analysis-to-document provenance.

4. Findings, limitations and relationship with DDTA

KerML directly falsifies the idea that Interaction should be a generic synonym for “edge”: an Interaction is explicitly both a Behavior and an Association. It suggests separating structural occurrence, behavior, relationship/connector and transfer semantics before deciding DDTA BAE classes. DDTA should reuse the lesson of semantic separation without importing the full KerML ontology.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
205	foundation	8.3.4.3 Structure	Structure and behavior should remain distinct semantic responsibilities.
214	foundation	8.3.4.5 Connector	Generic relation/connection semantics can be separated from behavioral interaction.
216	challenge	8.3.4.6 Behavior	Internal behavior is not reducible to an interaction between actors.
247	foundation	8.3.4.9 Flow	Flow has structured transfer semantics beyond a generic relation.
248	challenge	8.3.4.9 Interaction	Treating Interaction as a universal edge would distort established modeling semantics.

Pag.	Ruolo	Sezione	Interpretazione DDTA
297	challenge	8.4.4.10 Interactions Semantics	Interaction has a deliberately compound identity.
357	foundation	9.2.7.2.9 Transfer	Payload transfer can be represented independently of the structural connection medium.

Bibliography mining / follow-up

- SysML 2.0 Language Specification
- KerML Semantic Library
- Dori/Reinhartz-Berger reflective metamodel as a conceptual-modeling comparison

Governance note

Questa scheda è un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.